

Final Project (Week 6) :

Project Title:

Developing a Multi-Agent AI Agent System

Project Description:

In this final project, students will design and implement a fully functional AI agent system capable of performing complex, multi-step tasks. Unlike the mid-term project, which focuses on simple question answering, this project emphasizes orchestration of multiple AI processes and automation.

Objectives include:

1. **Data Retrieval:** The AI agent should gather data from various sources such as PDFs, Google Sheets, APIs, or databases.
2. **Data Processing & Summarization:** Use Large Language Models (LLMs) to process, analyze, and summarize the collected information efficiently.
3. **Automated Actions:** Implement automated outputs based on the processed data, such as sending emails, generating reports, or updating dashboards.
4. **Workflow Orchestration:** Design a pipeline that coordinates multiple AI tasks sequentially or in parallel, ensuring logical and accurate results.
5. **Presentation & Explanation:** Document the workflow and functionalities of the AI agent, then present the project via a face-cam video, demonstrating both the system in action and a clear explanation of the underlying architecture and logic.
6. **Testing & Evaluation:** Evaluate the system for reliability, accuracy, and efficiency, ensuring that the AI agent completes multi-step tasks successfully.

This project simulates real-world AI applications and demonstrates the power of AI agents in automating complex workflows, making it ideal for developing professional skills in AI system design.